NATIONAL INSTITUTE OF TECHNOLOGY KARNATAKA, SURATHKAL

Reduced-Parameter Water Quality Index Prediction for Data-Scarce River Systems

Evidence from 32,968 CPCB river monitoring visits, 2018–2021

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Only 15% of river visits can compute a full WQI

14.7%

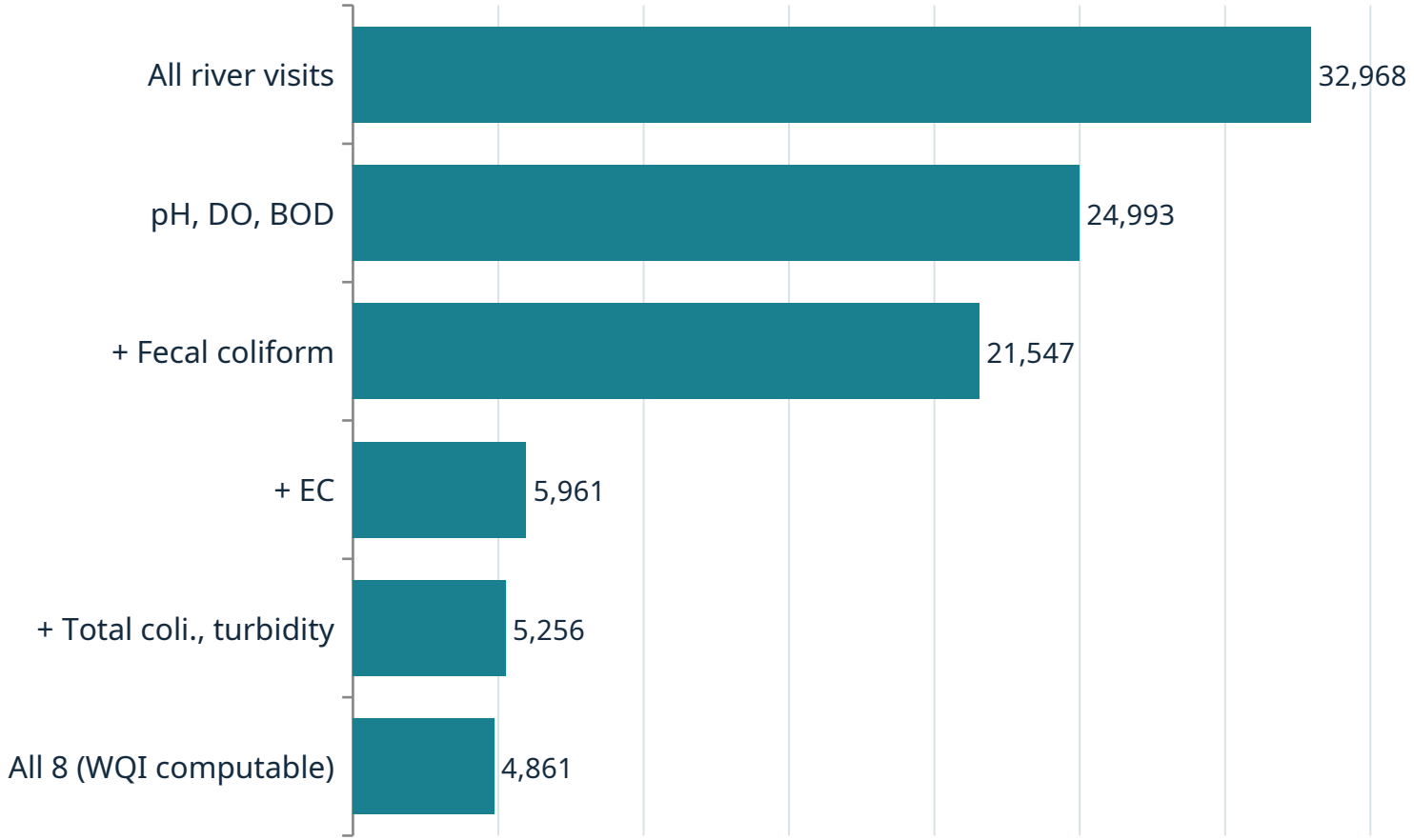
of 32,968 river visits report all 8 WQI parameters

75.8%

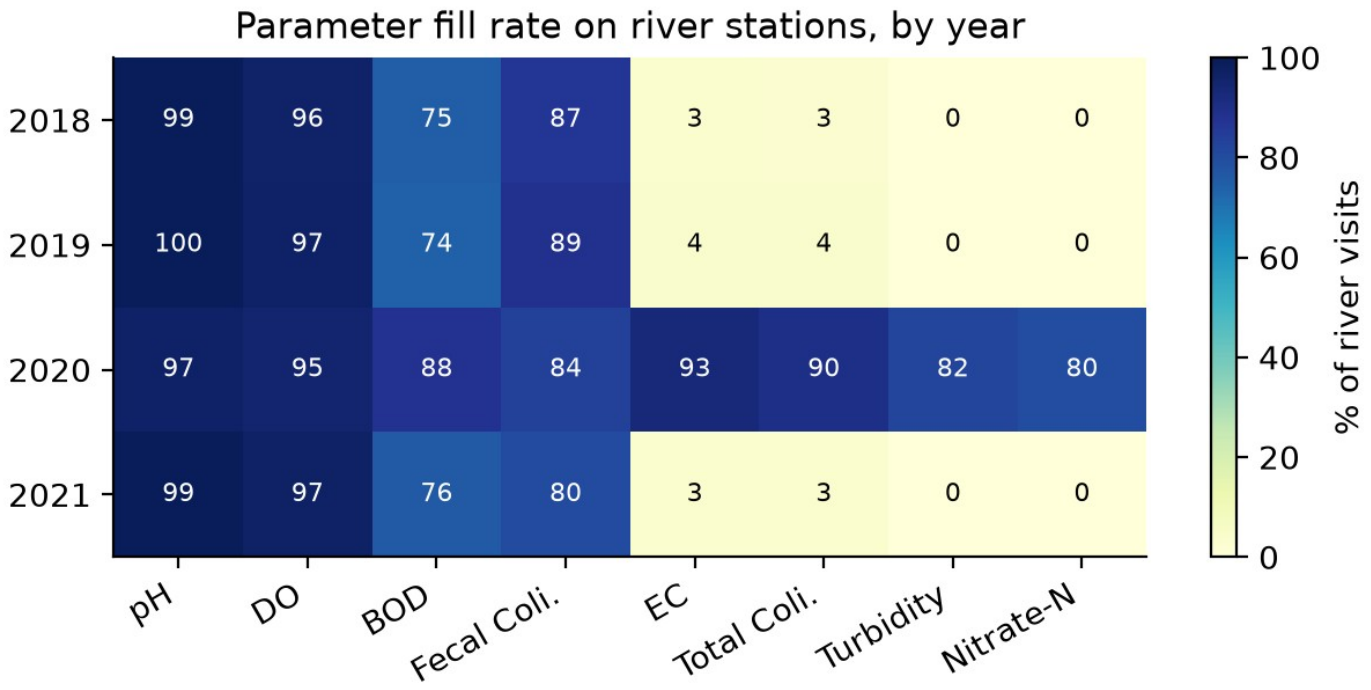
report pH, DO and BOD, the three best-covered parameters

Research question
Can reduced-parameter monitoring provide reliable WQI prediction for data-scarce river systems?

River visits reporting each nested parameter set



The full panel was recorded in one year only



Fill rate (% of river visits) by parameter and year

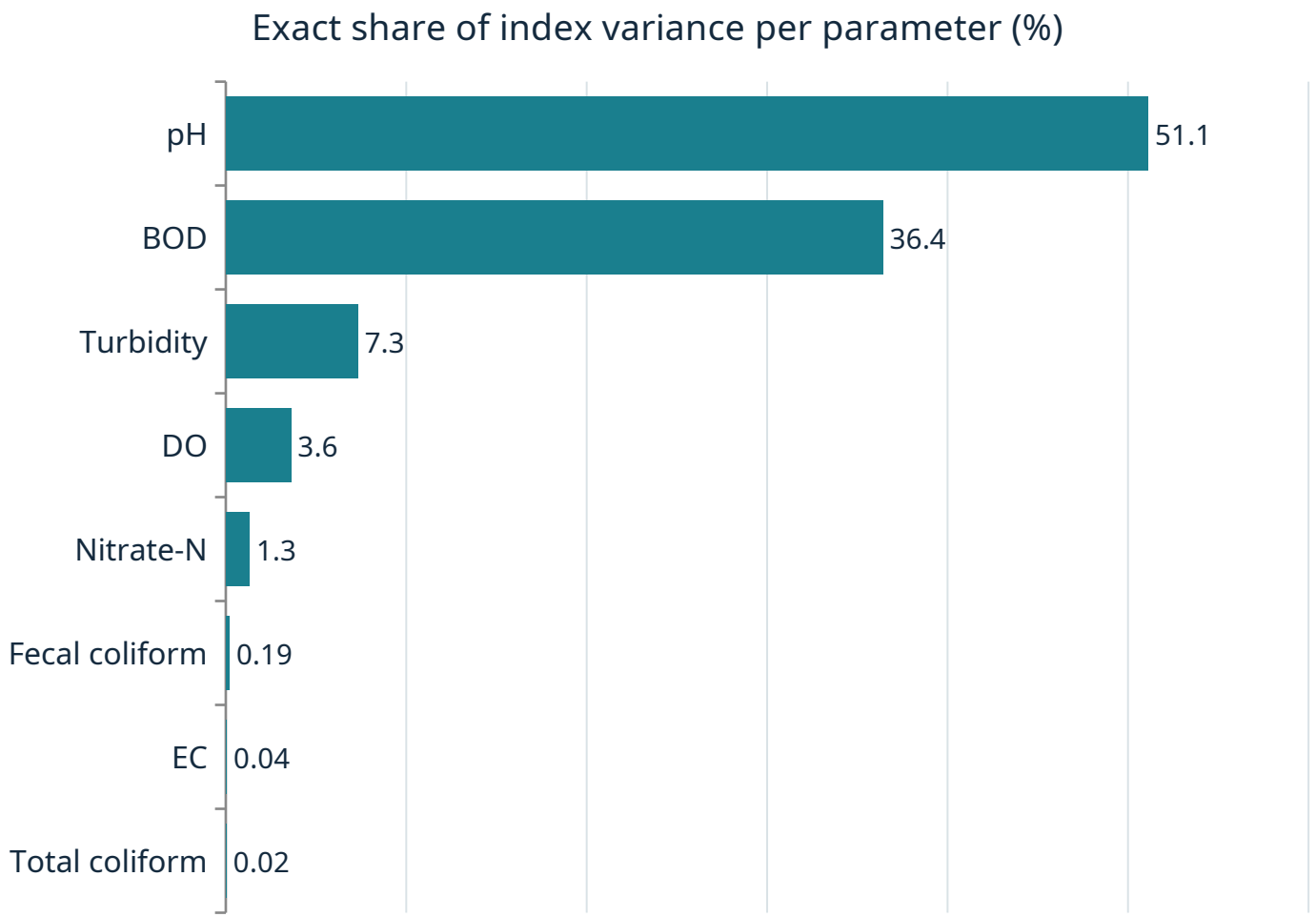
1 **211 raw CSVs**
physical, chemical and biological tables (57 MB)

2 **32,968 river visits**
888 stations, 87 rivers, joined and audited

3 **4,861 labelled visits**
593 stations with all 8 parameters, all in 2020

EC, coliforms, turbidity and nitrate were logged almost only in 2020. Other years carry a routine pH–DO–BOD–coliform panel: a real, not simulated, data-scarce setting.

pH and BOD drive 87.5% of the CPCB index



CPCB weighted-arithmetic WQI

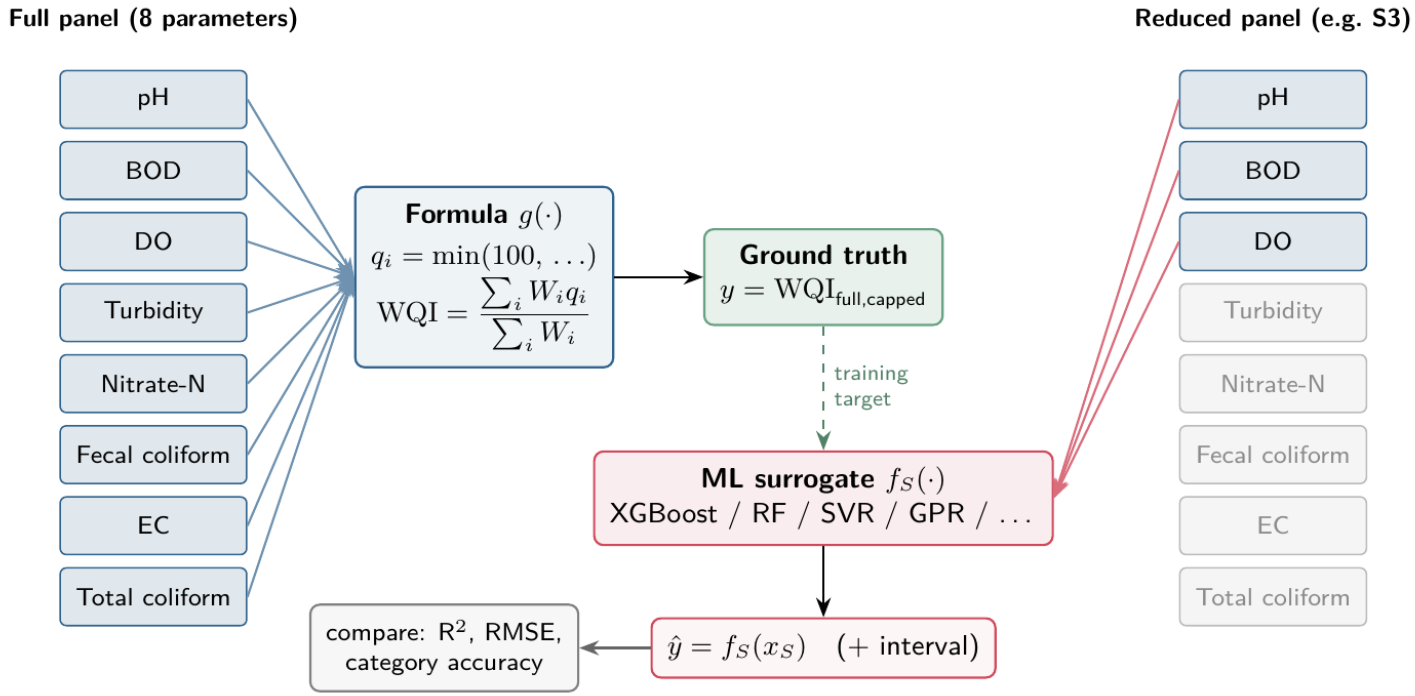
$WQI = \Sigma W_i \cdot q_i / \Sigma W_i$, with q_i capped at 100

Unit weight $W_i \propto 1 / S_i$ (S_i = standard)

Fecal coliform exceeds its bathing standard on 34% of visits, yet explains 0.2% of the index.

Weights $\propto 1/S_i$ reward parameters whose standard is a small number: 2 pH units or 3 mg/L, versus 500 MPN/100 mL.

Approach: learn the full-panel WQI from a few inputs



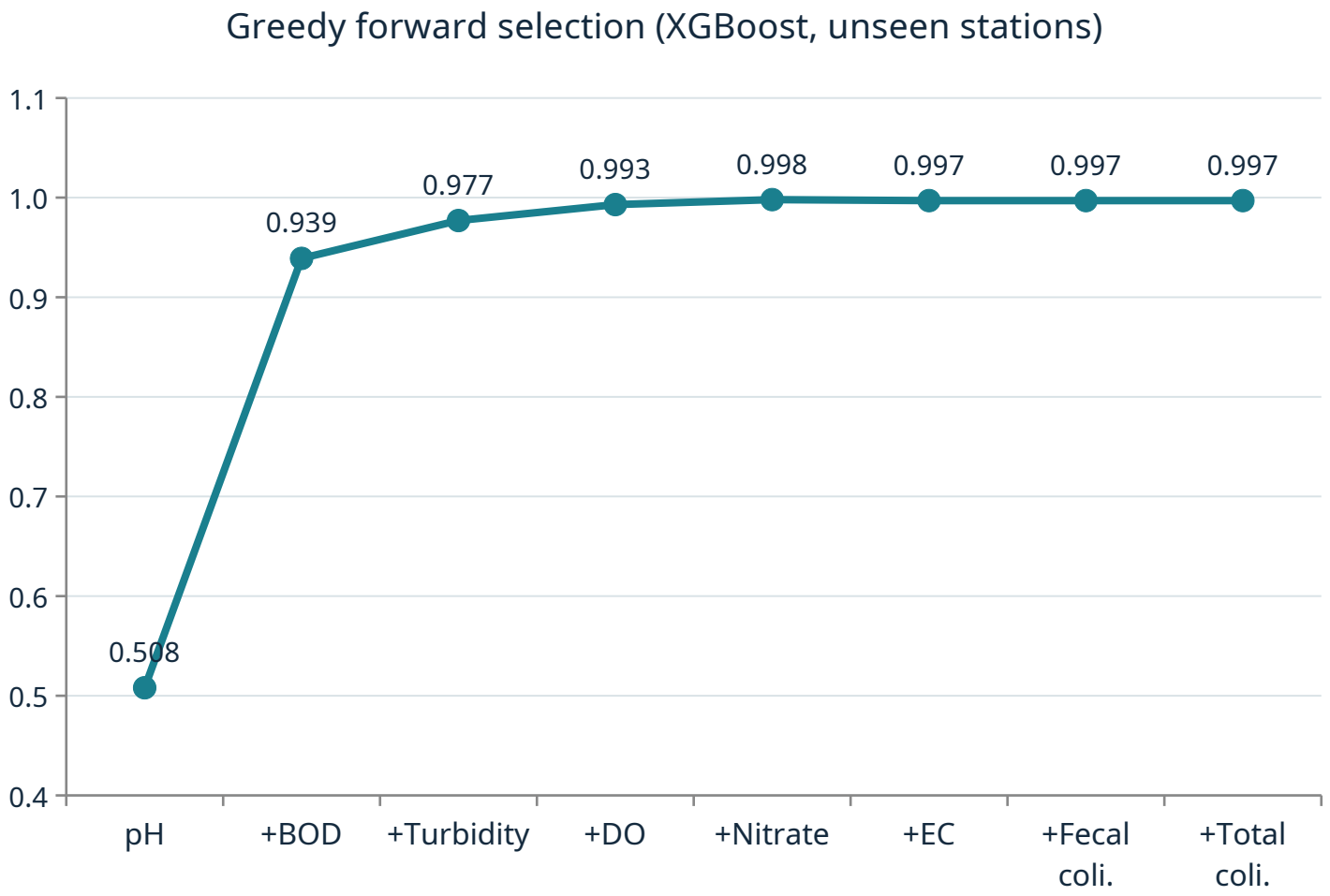
The target never changes: every model predicts the same full-panel WQI, so subset sizes are compared fairly.

8 model families
Formula (no ML), Linear, kNN, SVR, MLP, Random Forest, XGBoost, Gaussian process

12 named + all 255 subsets
3, 4, 5, 7 and 8 parameters; availability-, importance- and sensor-based

5 validation schemes
random rows, unseen stations (primary), unseen rivers, states, season

Importance: pH › BOD › turbidity › DO › nitrate



0.94

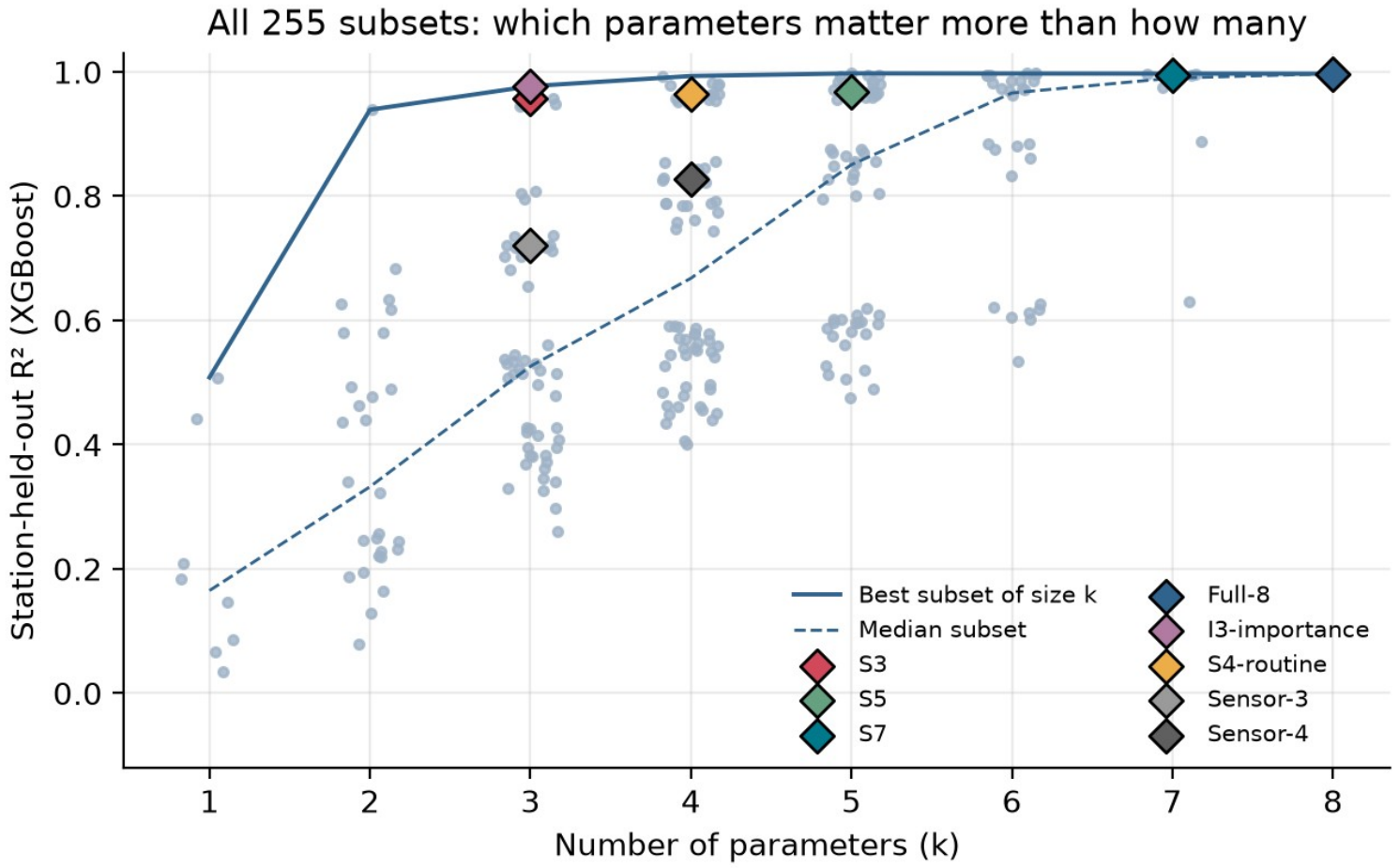
R^2 from pH and BOD alone

0

R^2 lost when coliforms or EC are removed from the full model

Correlation ≠ importance: the coliforms correlate with WQI ($\rho \approx 0.38$) only because sewage raises BOD too. Seven methods agree on the order.

Choosing the right parameters beats adding more



0.977

best 3-parameter set: pH, BOD, turbidity

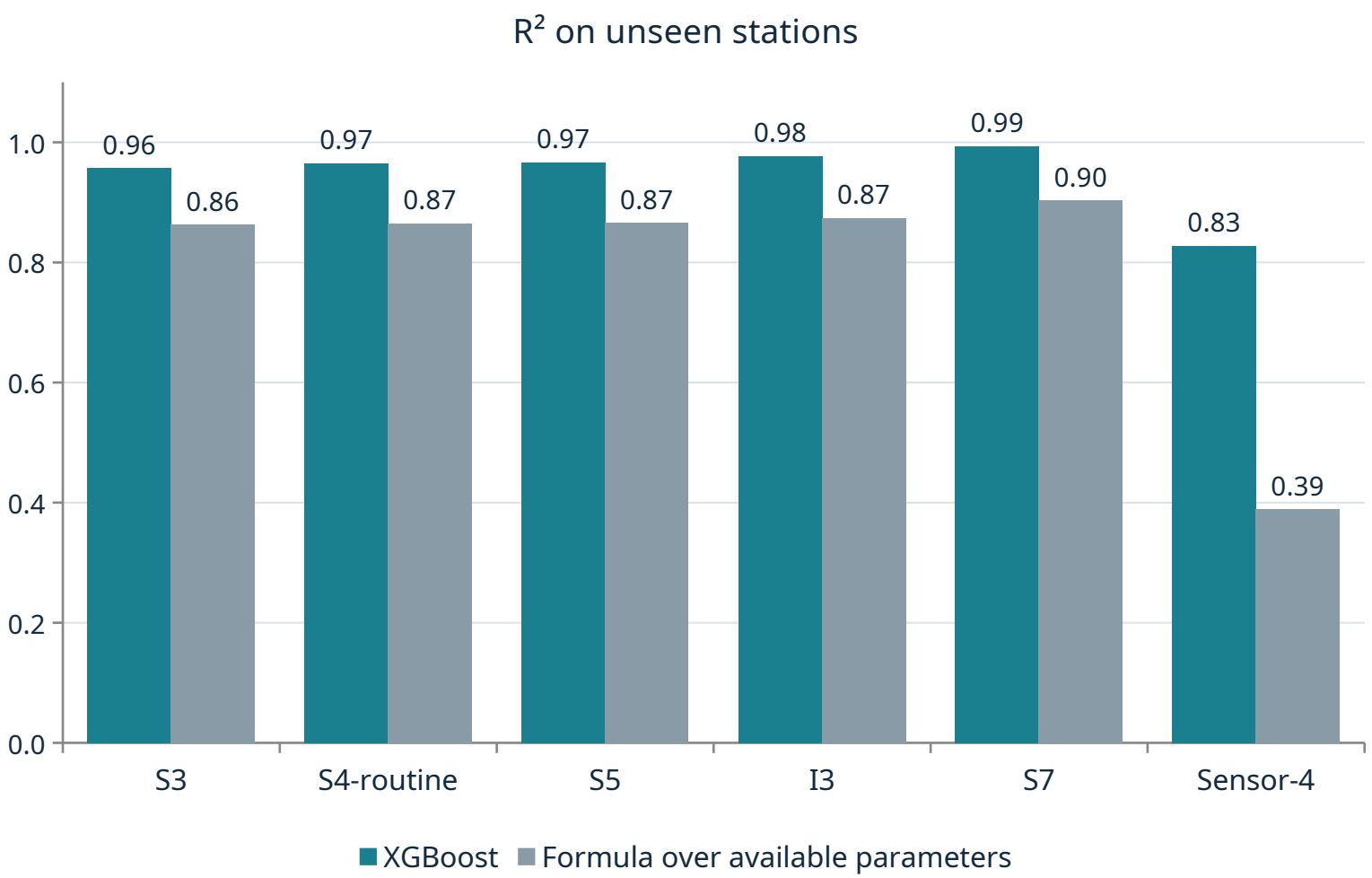
0.53

median 3-parameter set

2nd / 56

rank of the planned S3 (pH, DO, BOD): R^2 0.957

Machine learning clearly beats re-weighting the formula



- XGBoost is best for every subset; RF, MLP and GPR are close behind
- The subset matters far more than the model: for S3, flexible models span R^2 0.934–0.957
- Linear regression fails ($R^2 \approx 0.74$): the capped sub-indices are piecewise-linear
- The model learns the expected contribution of the missing parameters; the formula cannot

Three parameters predict the WQI at unseen stations

0.957

R^2 (NSE) for pH, DO, BOD

2.9

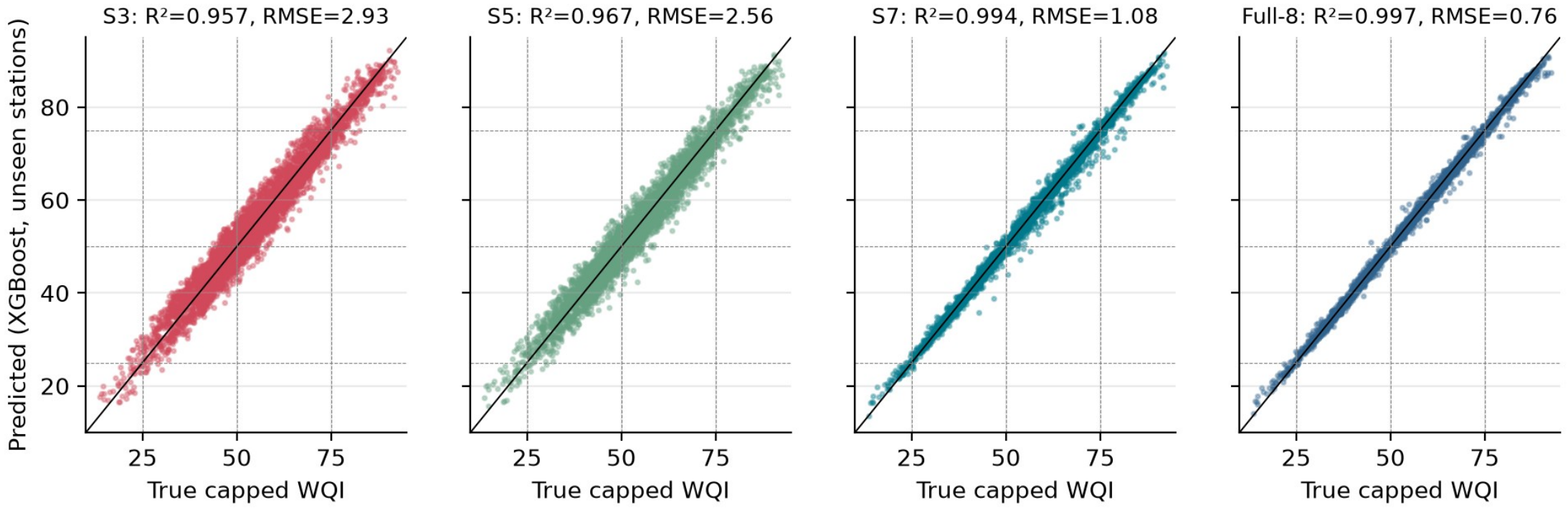
RMSE in WQI points (target s.d. 14.1)

90.6%

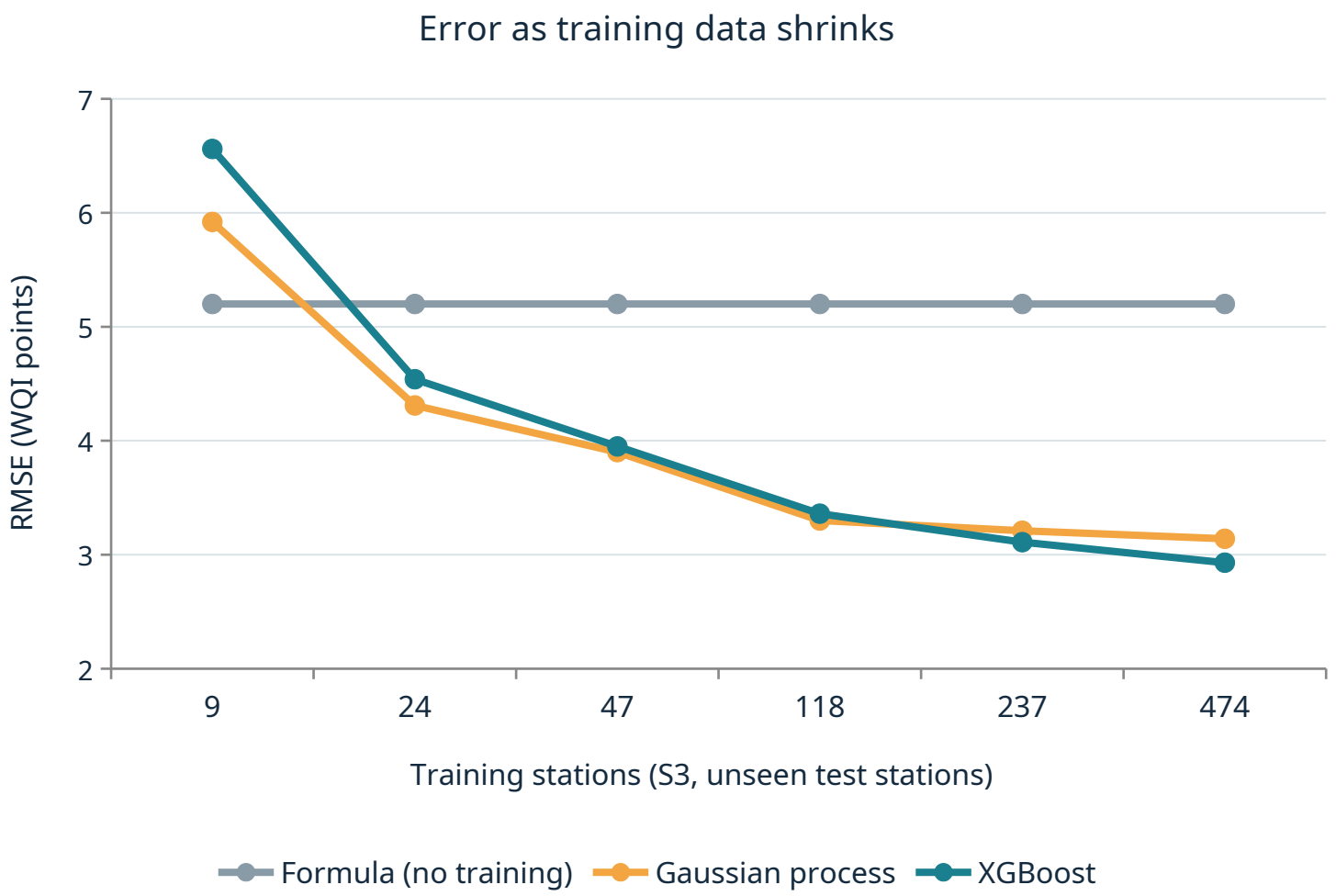
category accuracy (baseline 58%)

0

visits off by two or more categories



Models need ~25 training stations to beat the formula



1. **< 25 stations**
   use the formula

2. **25–100 stations**
   Gaussian process

3. **> 100 stations**
   XGBoost

Neural networks fail when data are scarce ($R^2 < 0$ with 9 stations). Extra parameters only pay off beyond about 50 stations.

Calibrated intervals; pH precision matters most

Split-conformal 90% intervals

90.3%

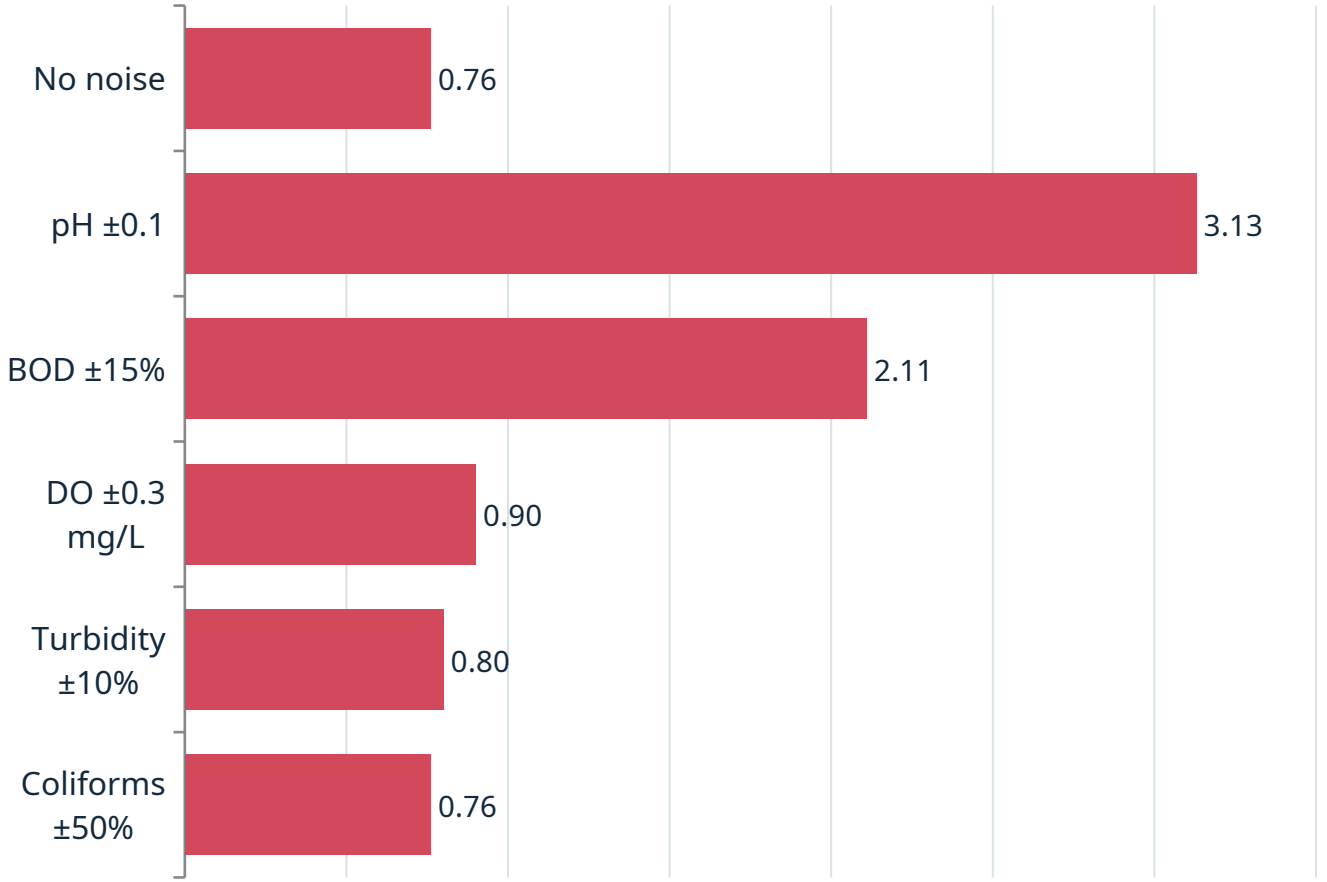
empirical coverage on unseen stations (S3)

± 4.7

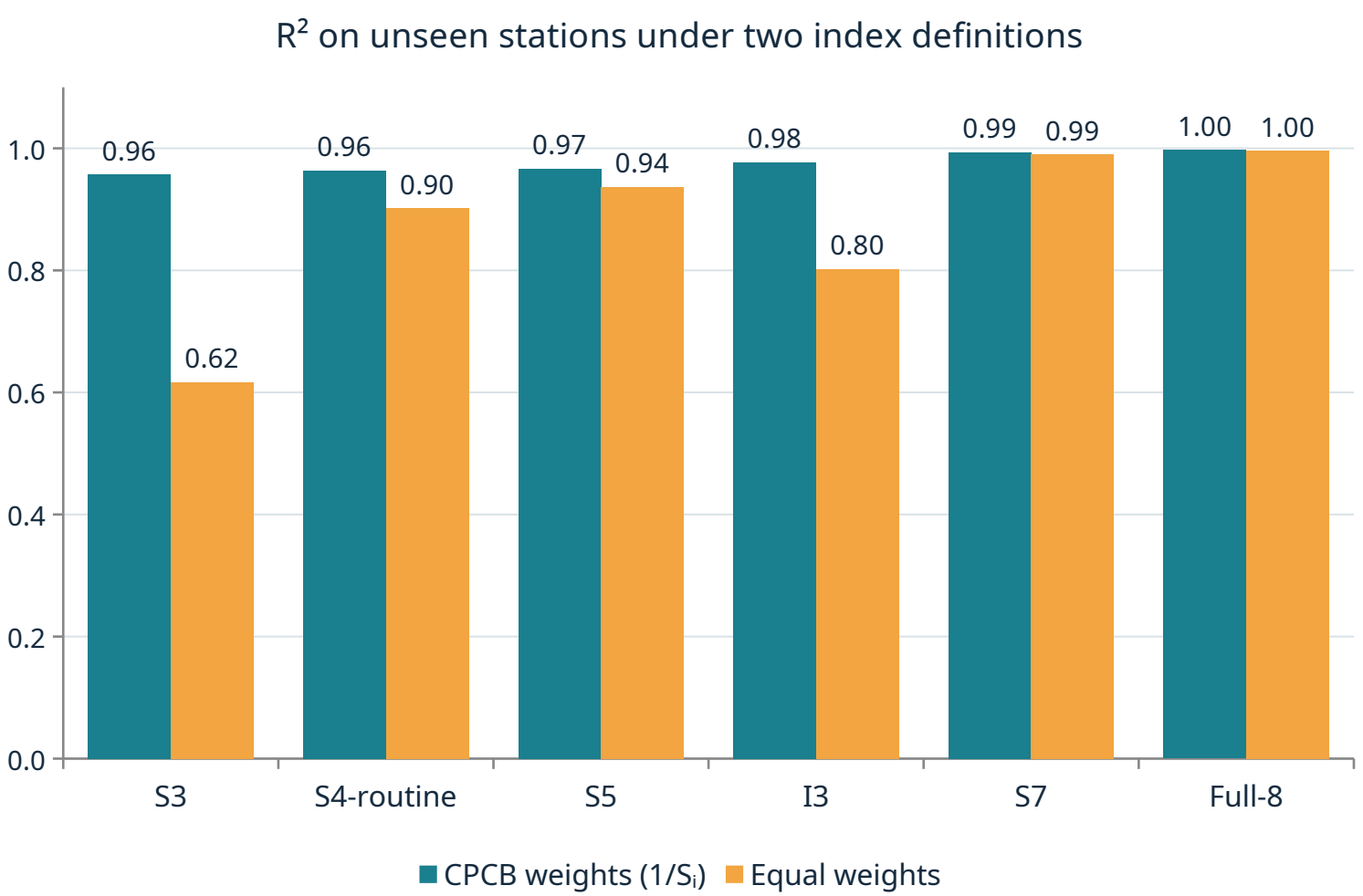
WQI points for S3; ± 1.6 for S7

62% of visits get an interval inside a single category; the rest are flagged “category uncertain”.

Full-panel model error with field noise on one parameter



Which panel works depends on the index definition



- Equal weights shift importance to the coliforms (52%) and turbidity (18%)
- Uncapped index: S3 R^2 0.37, but pH + BOD + turbidity reaches 0.98
- EC standard 250–4,000 µS/cm changes ≤ 0.35% of categories
- S7 is the only panel robust to every definition

Deployment: first full-panel WQI for 255 stations

20,132

visits reconstructed (2018–2021)

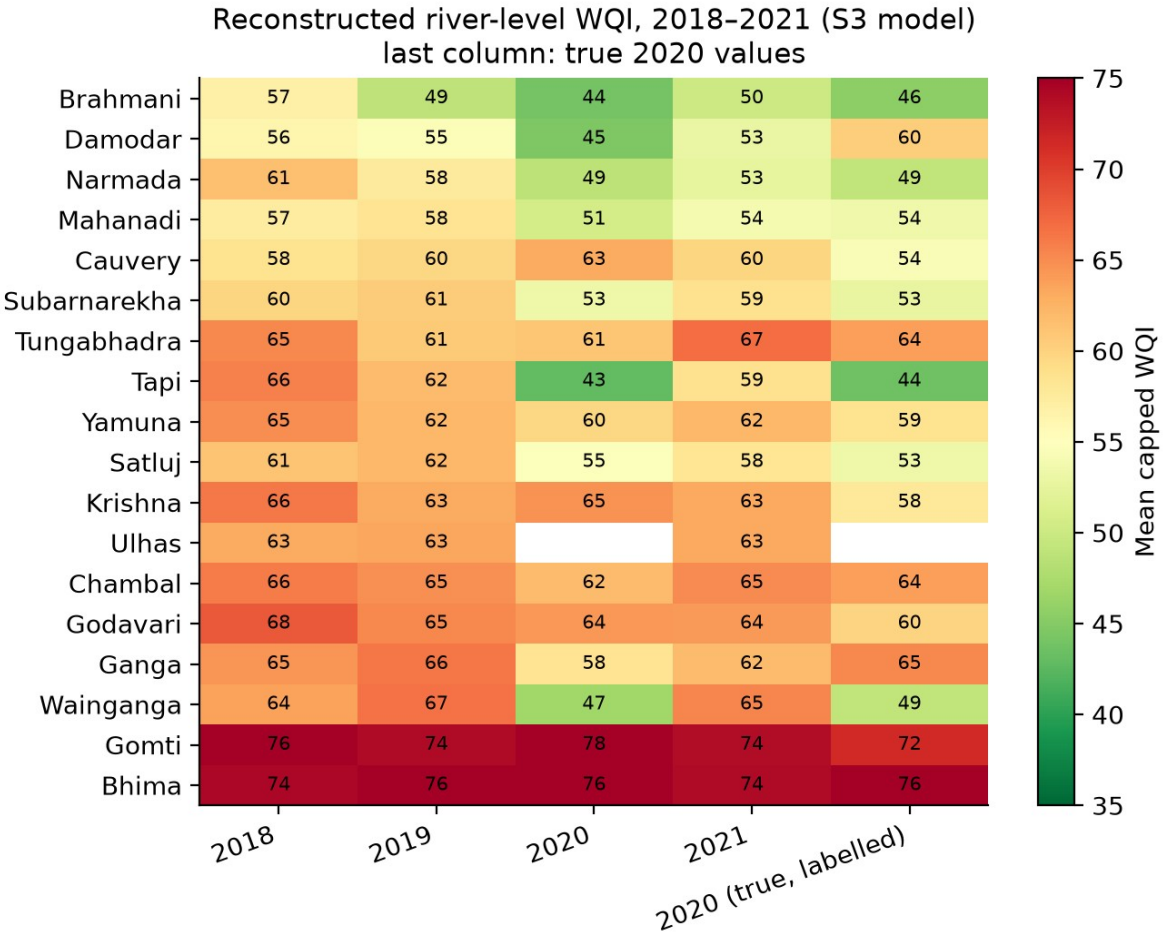
255

stations that never had a WQI

3.0

shift-corrected RMSE (vs 2.9 in CV)

Caveat: outside 2020, BOD ≤ 1 mg/L is never reported (left blank). Year-to-year trends must account for this reporting floor.



Recommended monitoring kit

Tier	Parameters	R^2	Category acc.	90% interval
0 Routine (no new cost)	pH, DO, BOD, fecal coliform + location, month	0.977	93.9%	± 3.6
1 + turbidity probe	pH, DO, BOD, turbidity	0.994	96.8%	± 1.6
2 + nitrate field kit	pH, DO, BOD, turbidity, nitrate-N	0.998	97.7%	± 1.1
Public-health robust	S7: + both coliforms, EC	0.994	97.0%	± 1.5
Screening only	pH, DO, EC, turbidity (all probes)	0.828	82.4%	± 9.8

1. Calibrate pH to ±0.05 and quality-control BOD
2. Route to a dedicated model for whatever was measured
3. Publish the conformal interval with every estimate
4. Report fecal coliform exceedance alongside the WQI

Answer: yes, with the right three parameters

1. **pH, DO and BOD reproduce the CPCB WQI**
   R^2 0.957 and 91% category accuracy at unseen stations; 0.994 with a turbidity probe

2. **Learning beats re-weighting**
   once data from about 25 stations are available, with calibrated ±4.7-point intervals

3. **Mind the index and the instruments**
   the CPCB weights ignore coliforms, and pH precision matters more than parameter count

Next: more full-panel years, censored-BOD handling, NSF/CCME indices, satellite turbidity, a field tool.

Thank you. Questions?